

Table 5. Characteristics of studies included—preclinical studies using Wharton’s-Jelly- or umbilical-cord-derived MSCs ($n = 9$).

Authors	Animal Model (n)	Burn Depth	Cell Species	Administration Method, Medium	Cell Dosage		Results in MSC Group
					Dose (Passage)	Cells/cm ² ¹	
Afzali et al. (2022) [75]	Rat (40)	Superficial partial thickness	Xe UC-MSC, human	Cell scaffold and local injection, PRP cryogel and cell culture medium (two groups)	2×10^6 cells (n/a)	n/a	Improved wound healing, increased wound closure rate, best results in scaffold group. Increased re-epithelialization and increased early neo-angiogenesis.
Cheng et al. (2020) [76]	Porcine (4)	Full thickness	Xe WJ-MSC, human	Topical application, in situ fibrin–HA bioink	1×10^6 cells/mL (P1)	n/a	Better healing with less inflammation, scarring and contraction. Increased re-epithelialization, better archeology. No infection.
Gholipour-Kanani et al. (2012) [77]	Rat (12)	Full thickness	Xe WJ-MSC, human	Cell scaffold, Cs:PVA nanofibrous web	4×10^4 cells/scaffold (P1)	1.8×10^4	Accelerated wound healing and wound closure rate. Less inflammation. Increased re-epithelialization and granulation, regular pattern of regenerated collagen.
Gholipour-Kanani et al. (2014) [78]	Rat (12)	Full thickness	Xe WJ-MSC, human	Cell scaffold, PCL:Cs:PVA nanofibrous web	4×10^4 cells/scaffold (P1)	4.2×10^4	Accelerated healing process, but longer than non-burn wound group. Increased collagen deposition, granulation, and re-epithelialization. No complications reported.
Hashemi et al. (2020) [79]	Rat (32)	Full thickness	Xe WJ-MSC, human	Cell scaffold, HAM	1×10^6 cells/scaffold (P3)	n/a	Increased rate of healing, re-epithelialization, granulation. Mature and organized scar tissue, less hemorrhage and inflammation.
Jehangir et al. (2022) [80]	Rat (35)	Partial thickness	Xe WJ-MSC, human	Cell scaffold, A-PCL composite scaffold and collagen (two groups)	1×10^5 cells/cm (P1)	1×10^5	Increased wound healing and complete epithelialization in both MSC groups, best in A-PCL-WJ-MSC group with complete epidermal restoration and near normal skin appendage